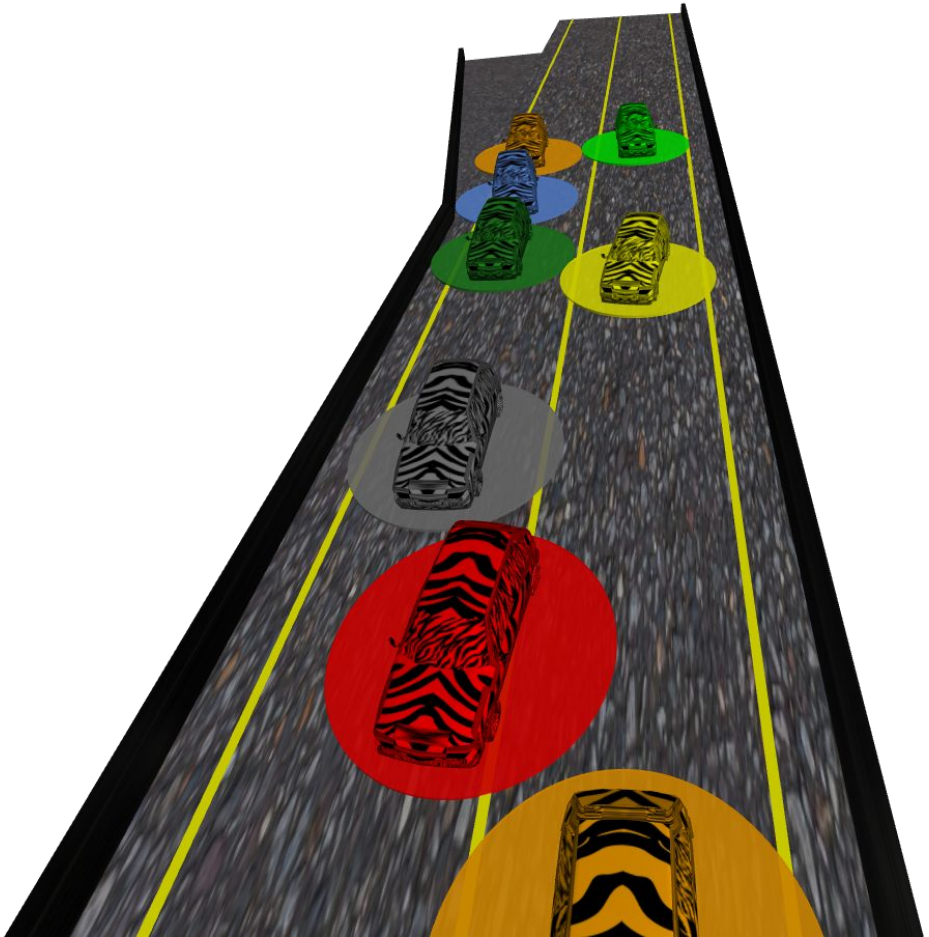
ALGAMES

A Fast Solver for Constrained Dynamic Games

Simon Le Cleac'h , Mac Schwager and Zachary Manchester

OUTLINE

- Motivation - Objective
- Dynamic Games
- ALGAMES Solver
- Results
- Future Work



Robotic Exploration Lab

Motivation

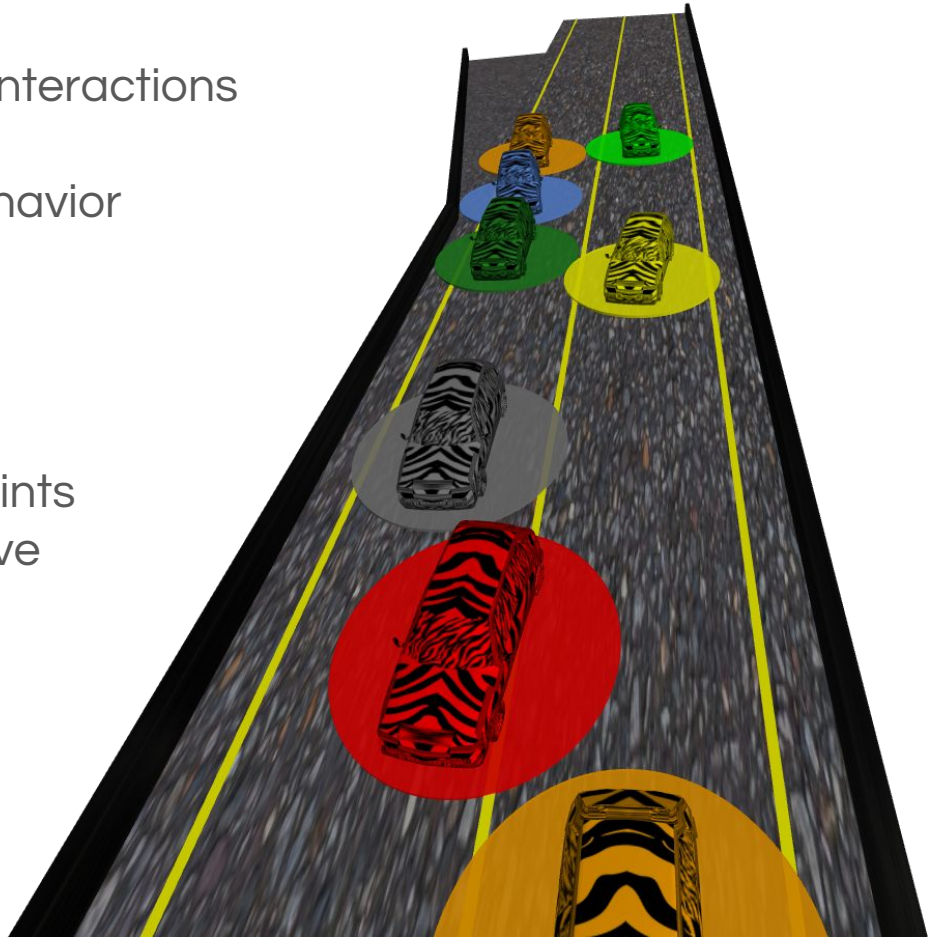
- Usual approach
  - Prediction / Planning
  - Decoupling
  - Freezing robot

- Our approach
  - Maintain coupling
  - Game between cars

OBJECTIVE

- Complex multiplayer interactions
- Maintaining coupling
- Rich autonomous behavior

- Our approach
  - Fast solver
  - Handling constraints
  - Individual objective



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DYNAMIC GAMES

- Formulation
  - Dynamic Game
  - 2 players

$$x_{t+1}^{(1)} = Ax_t^{(1)} + Bu_t^{(1)}$$ Dynamics player 1

$$x_{t+1}^{(2)} = Ax_t^{(2)} + Bu_t^{(2)}$$ Dynamics player 2

$$||x_t^{(1)} - x_t^{(2)}|| < r$$ Collision

$$J^{(1)}(X^{(1)}, U^{(1)})$$ Objective player 1

$$J^{(2)}(X^{(2)}, U^{(2)})$$ Objective player 2

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DYNAMIC GAMES

- 2 players
  - There is no “optimal” solution.

Start 1 1

Start 2 2

2 End 2

1 End 1

Robotic Exploration Lab

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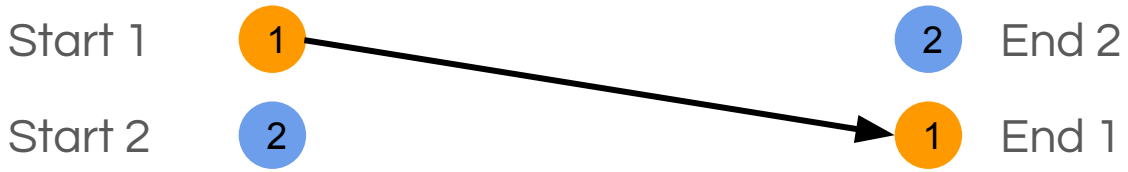

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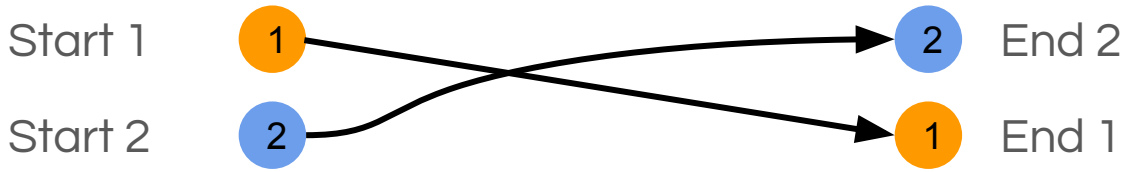

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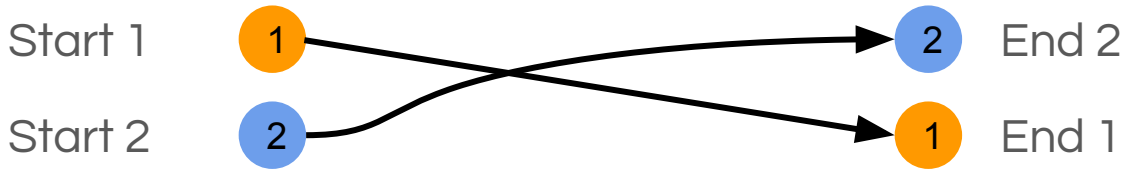

- Player 1 arrives first
- Player 2 arrives 2nd

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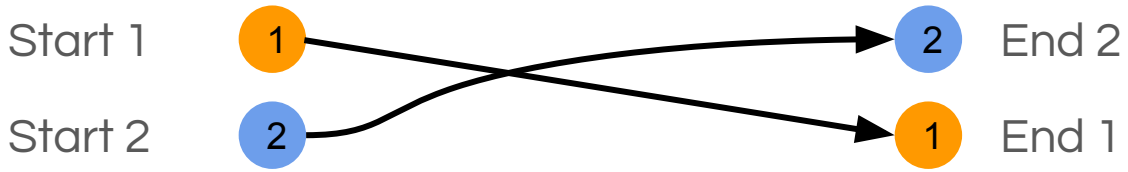

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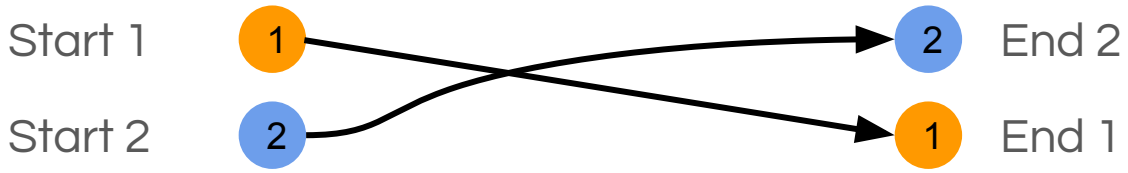

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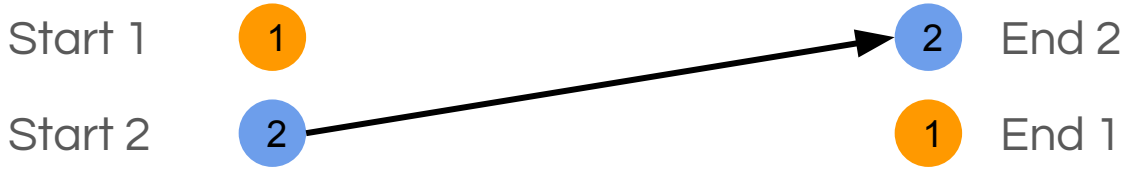

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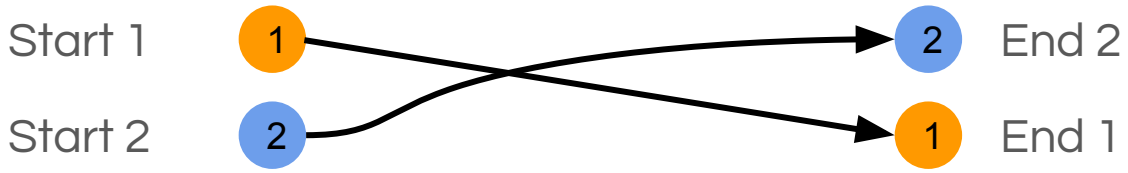

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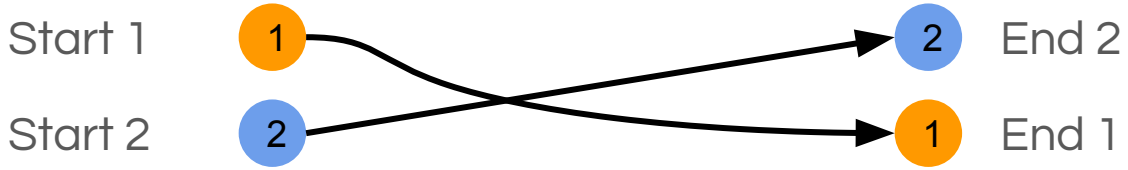

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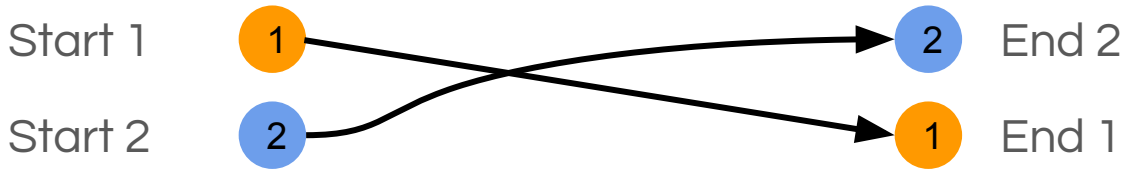

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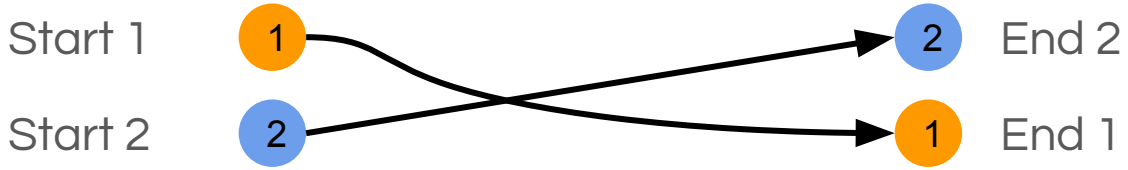

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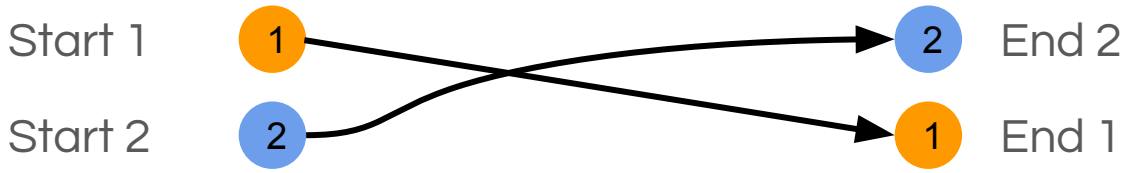

- Player 1 arrives first
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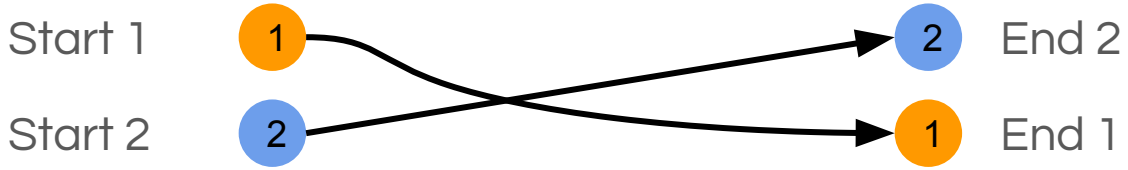
Cannot be optimal for both players at the same time.



- Player 1 arrives first
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DYNAMIC GAMES

- Nash equilibrium
  - One type of “optimal” solution
  - No cooperation
  - No hierarchy between players

- The idea is that no player can improve its cost by changing unilaterally its strategy.

DYNAMIC GAMES

- The idea is that no player can improve its cost by changing unilaterally its strategy.
  - Player 1's strategy is best response to player 2's strategy
  - Player 2's strategy is best response to player 1's strategy

- Formally

$$X^{(i)}, U^{(i)} = P^{(i)}$$ Strategy of player i

$$J^{(1)}(X^{(1)}, U^{(1)}, X^{(2)}, U^{(2)}) = J^{(1)}(P^{(1)}, P^{(2)})$$ Cost of player 1

$$J^{(2)}(P^{(1)}, P^{(2)})$$ Cost of player 2

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DYNAMIC GAMES

- The idea is that no player can improve its cost by changing unilaterally its strategy.
  - Player 1's strategy is best response to player 2's strategy
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- Formally

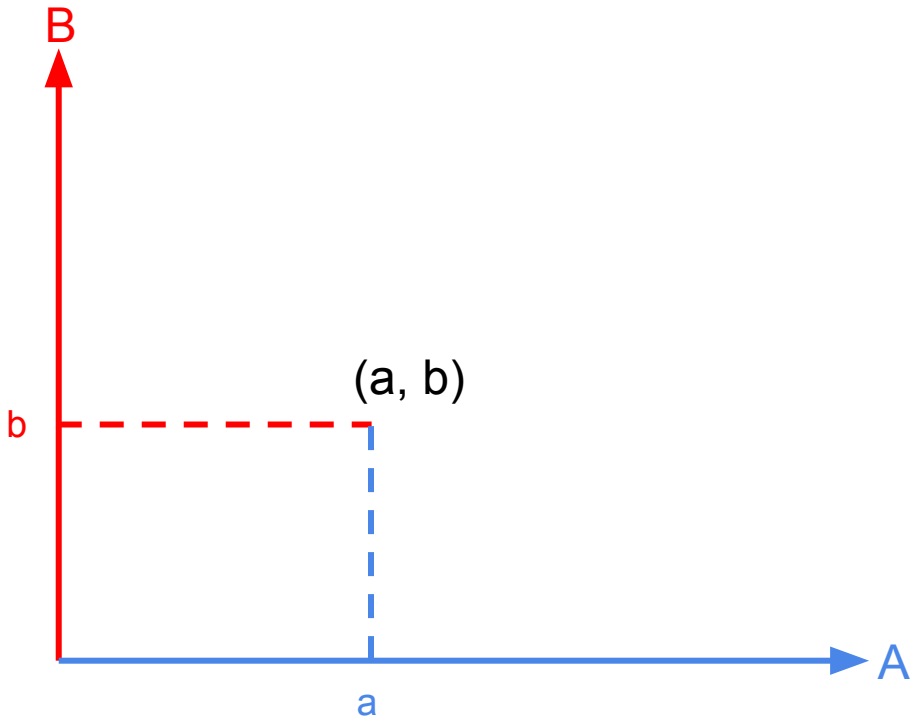
$$(P^{(1)*}, P^{(2)*})$$ Nash equilibrium

$$P^{(1)*} = min_{P^{(1)}} J^{(1)}(P^{(1)}, P^{(2)*})$$ Best response of 1

$$P^{(2)*} = min_{P^{(2)}} J^{(2)}(P^{(1)*}, P^{(2)})$$ Best response of 2

DYNAMIC GAMES

Generalized Nash Equilibrium example



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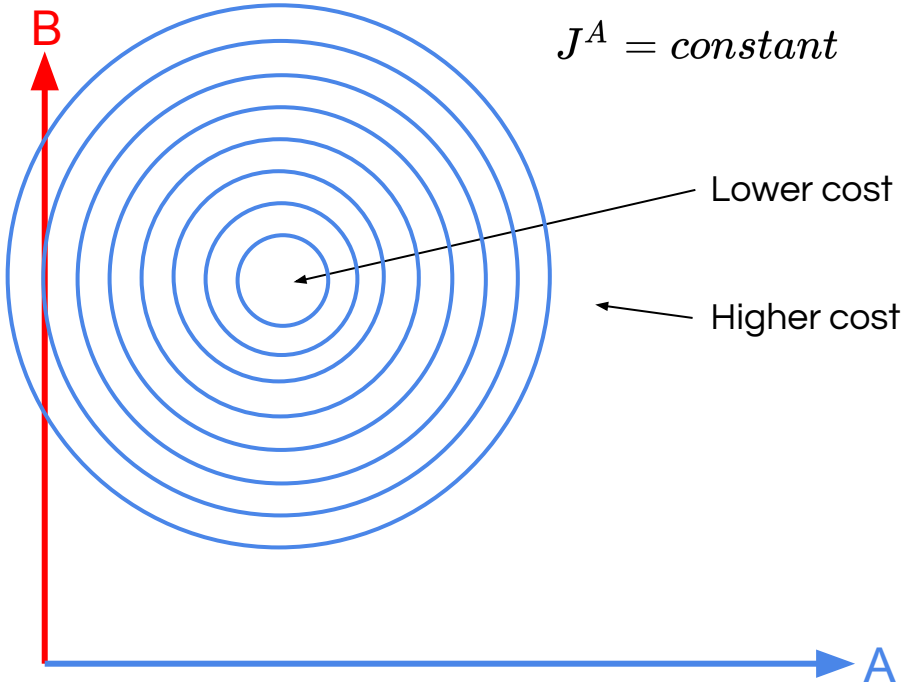

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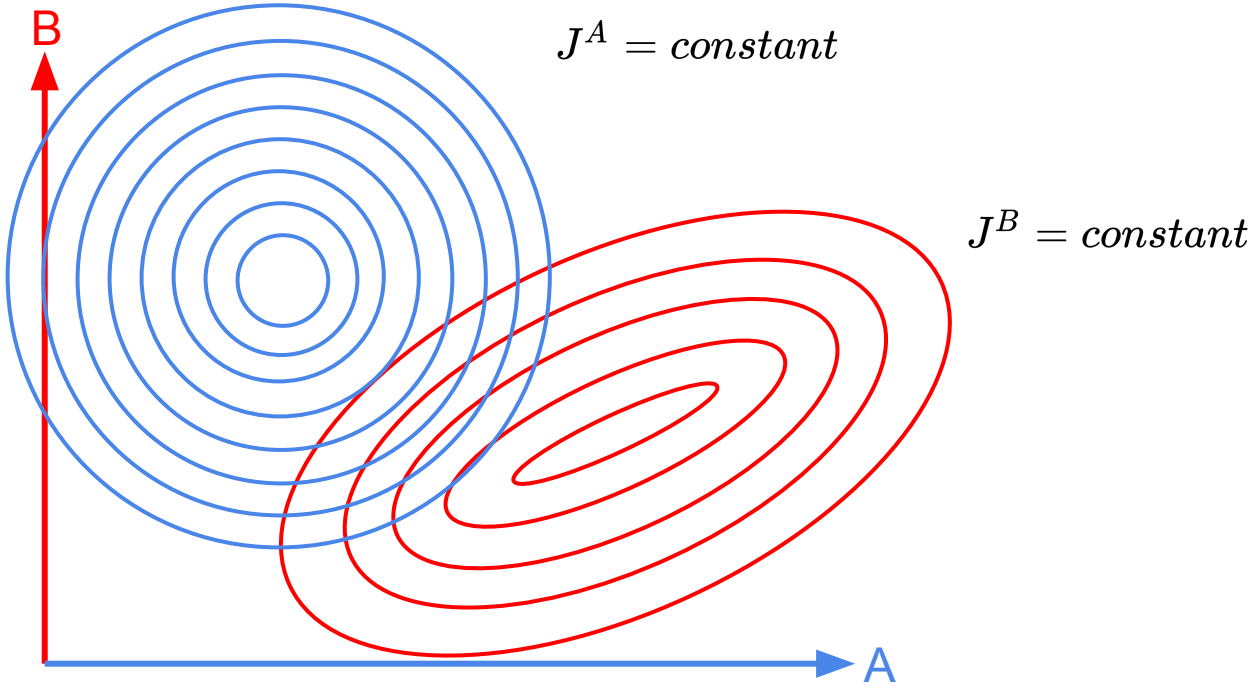
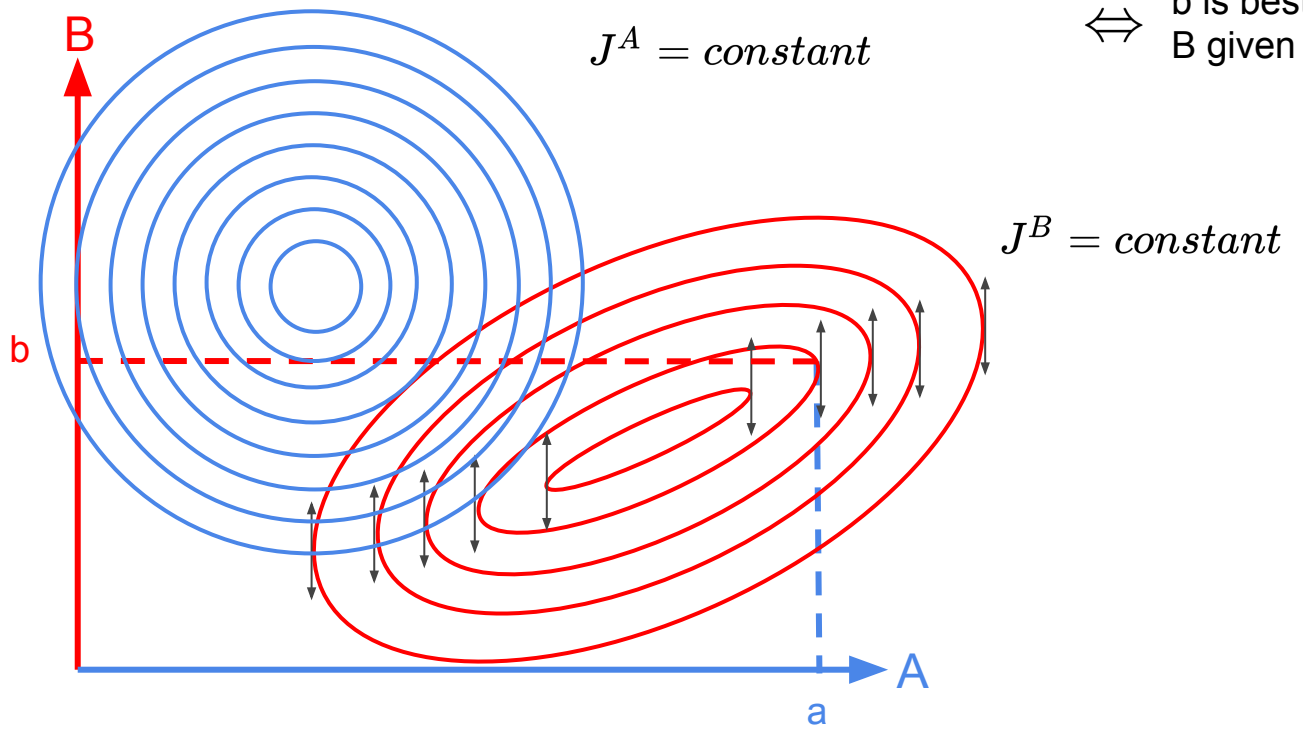


DYNAMIC GAMES

Generalized Nash Equilibrium example

$$\updownarrow \Leftrightarrow \nabla_b J^B(a,b) = 0$$

$\Leftrightarrow$ b is best response of B given a



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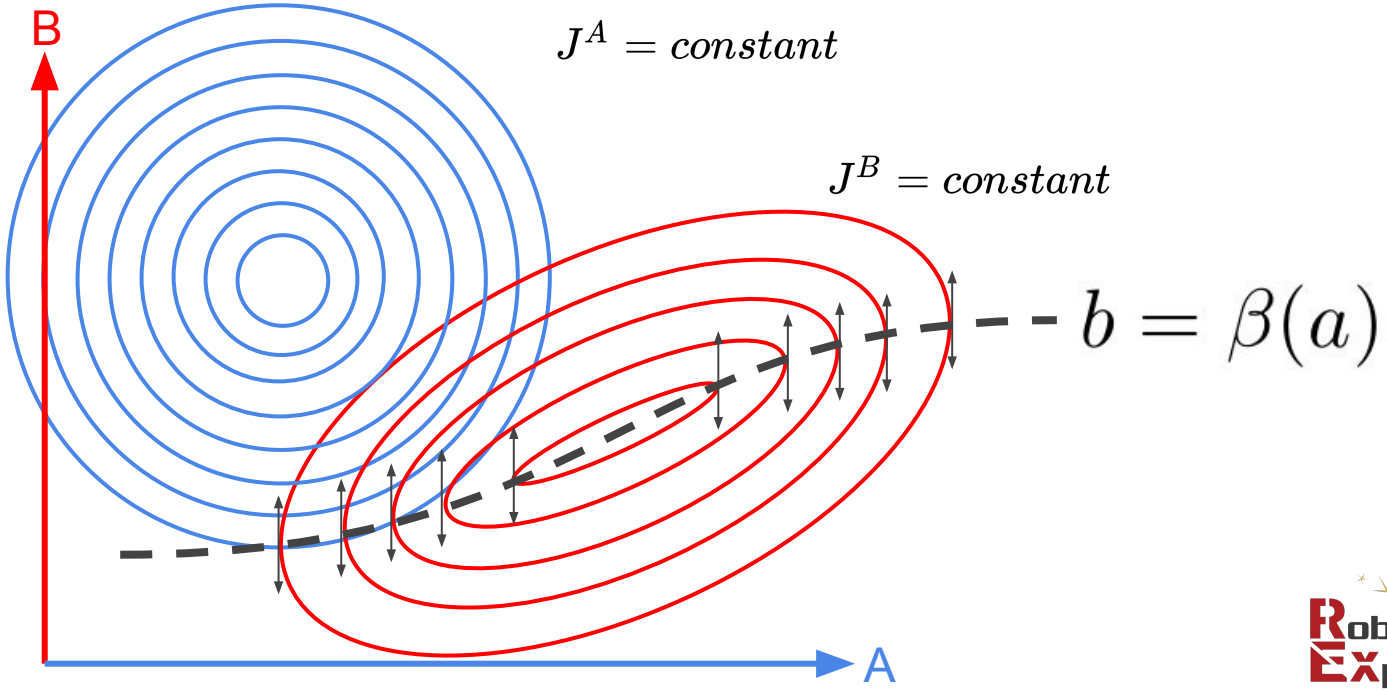

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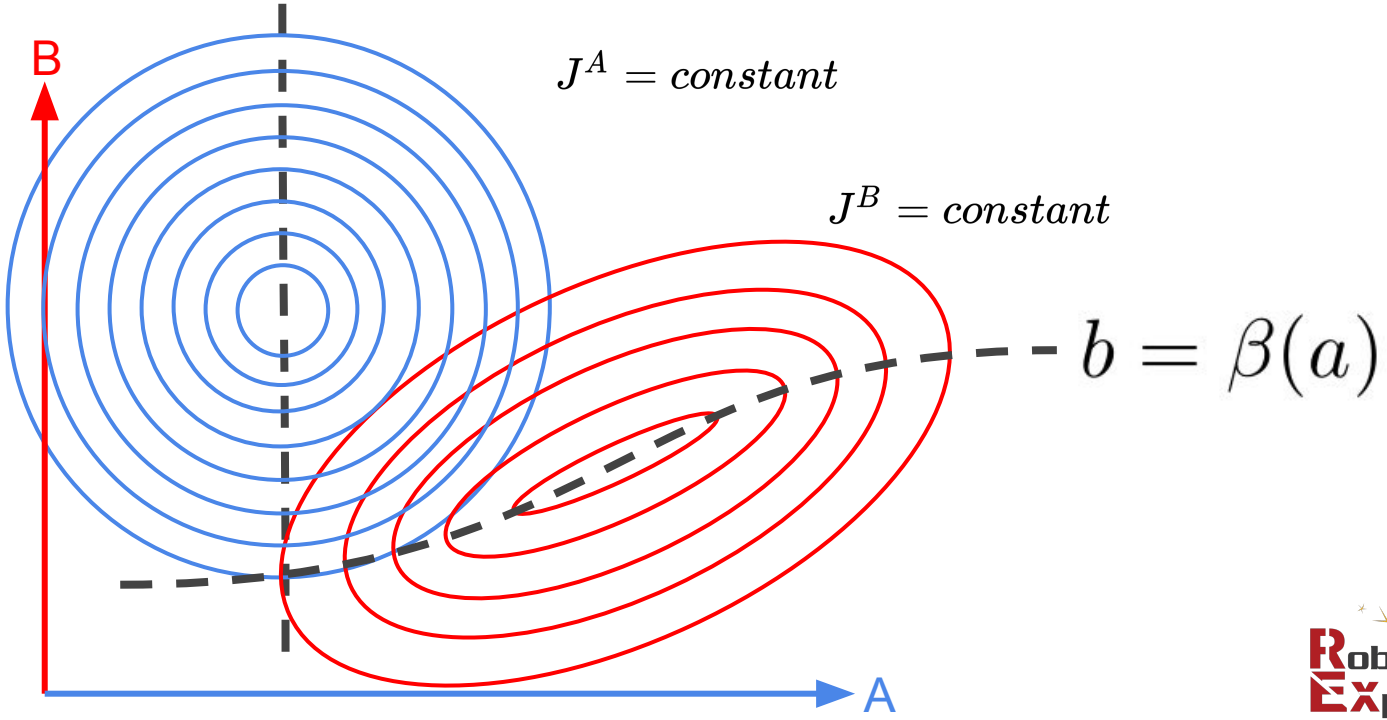

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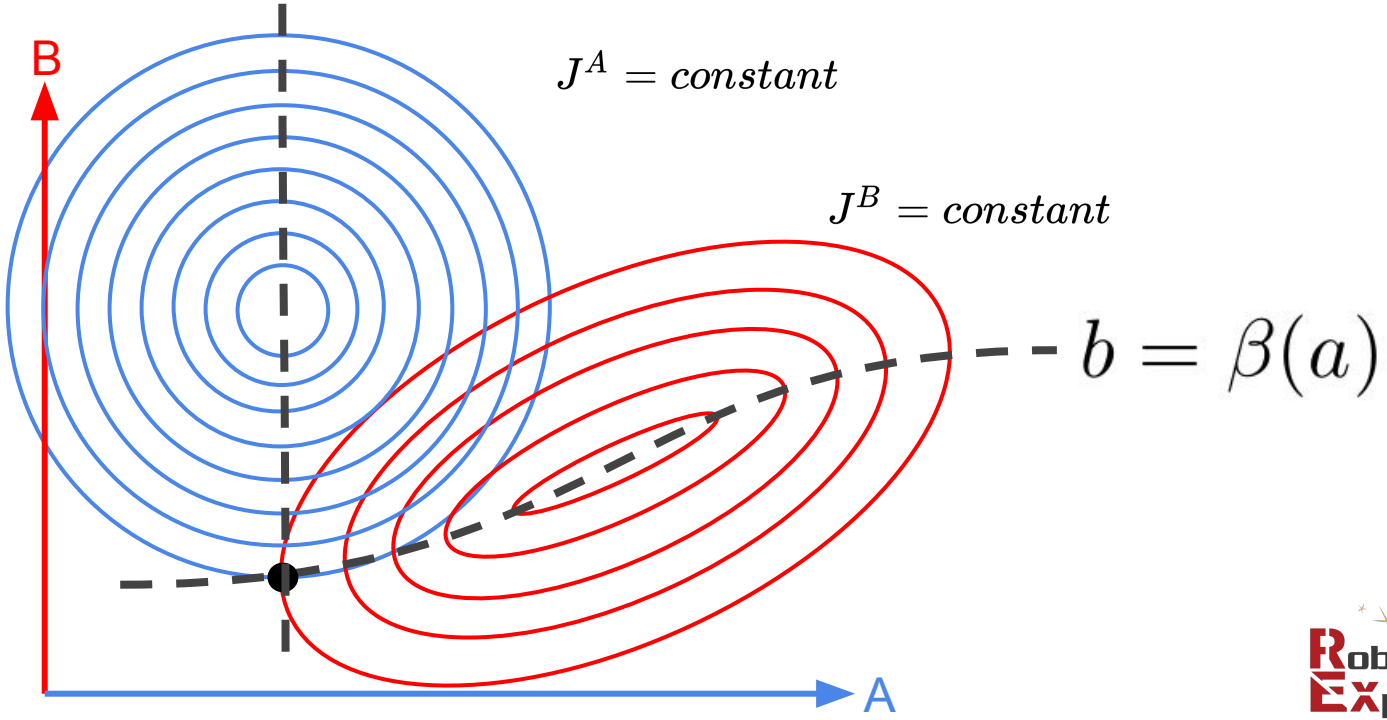

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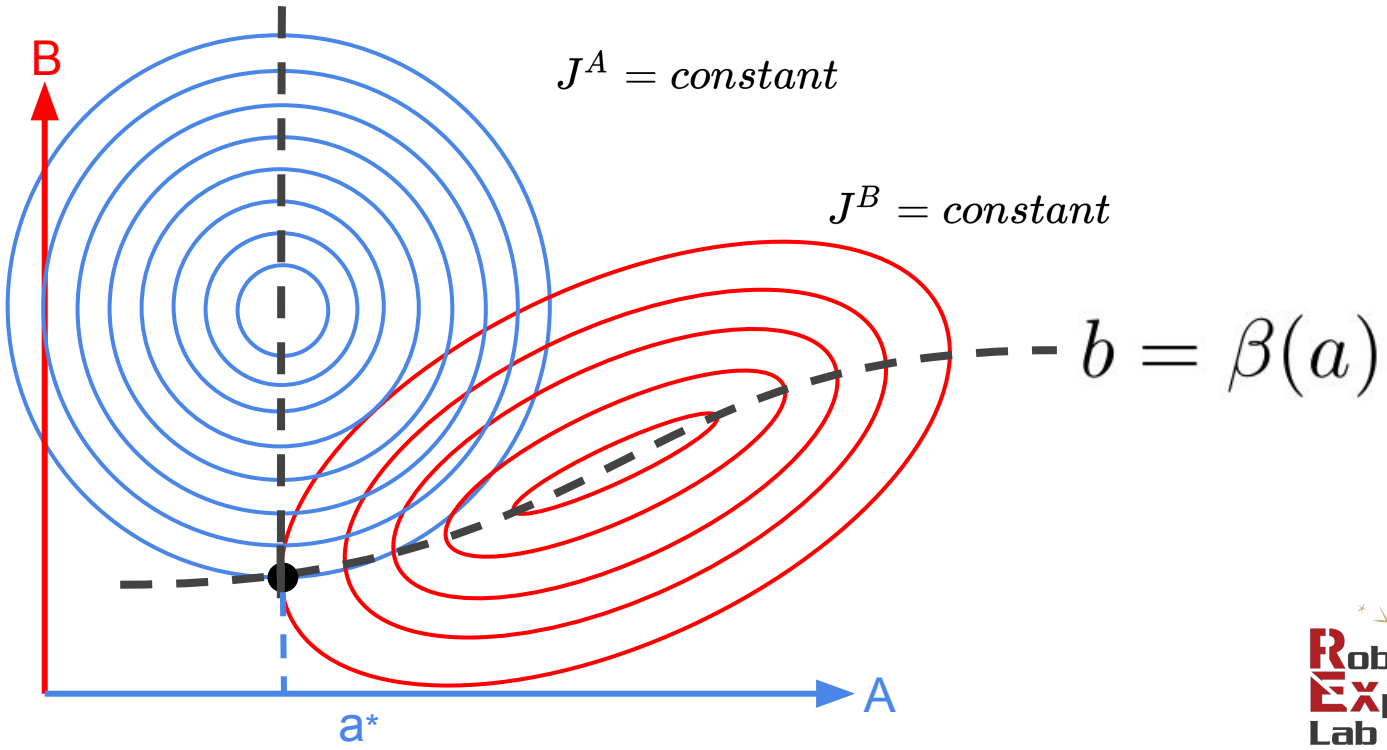

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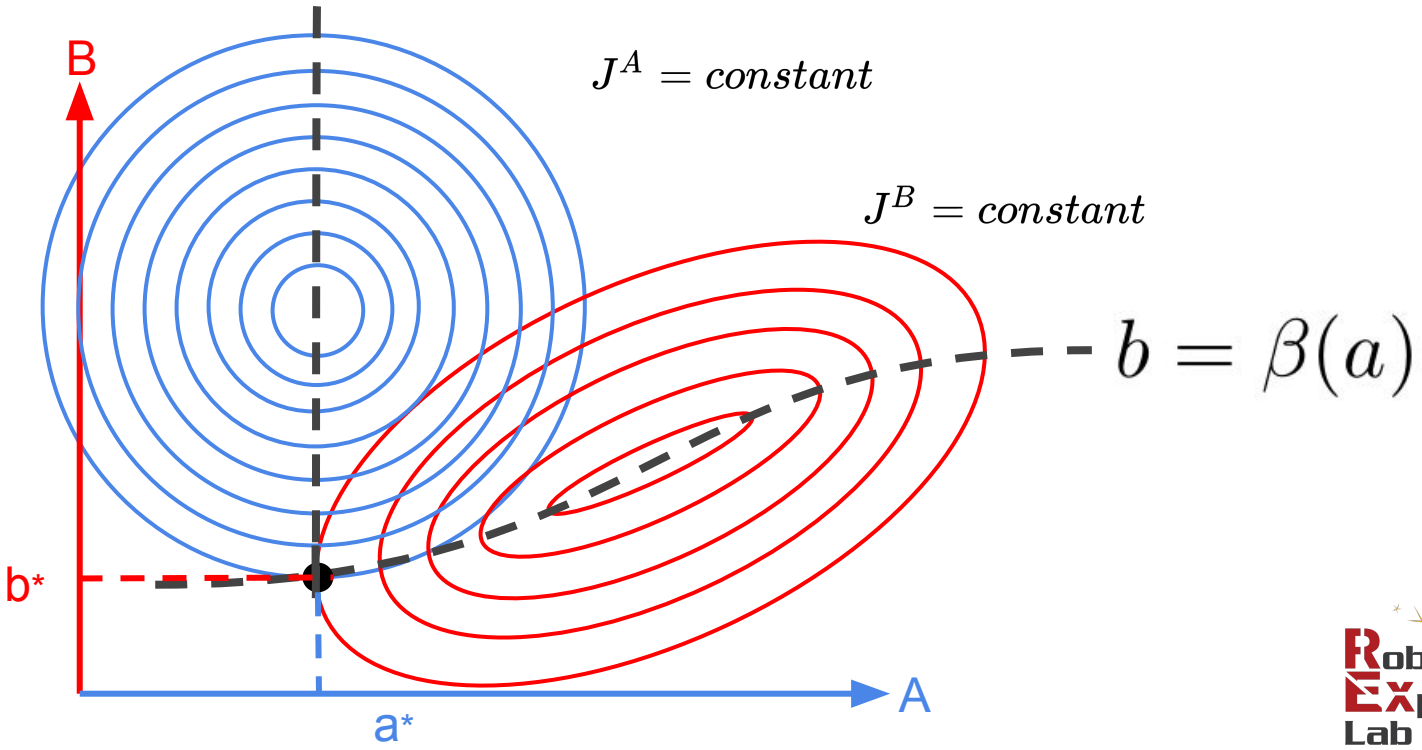

SOLVER

- Objective
  - General solver for dynamic games.
  - Nash equilibrium strategies.
  - Handle constraints.

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- Optimization problem

$$
\begin{aligned}
min_{P^{(i)}} \quad & J^{(i)}(P^{(1)}, P^{(2)}) \\
s.t. \quad & C^{(i)}(P^{(1)}, P^{(2)}) \leq 0
\end{aligned}
$$

SOLVER

- Objective
  - General solver for dynamic games.
  - Nash equilibrium strategies.
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- Optimization problem

$$min_{P^{(i)}} \quad J^{(i)}(P^{(1)}, P^{(2)})$$

$$s.t. \qquad C^{(i)}(P^{(1)}, P^{(2)}) \leq 0$$

- Augmented Lagrangian

$$L^{(i)}(P^{(1)}, P^{(2)}) = J^{(i)} + \lambda^{(i)^T} C^{(i)} + \frac{1}{2}\rho^{(i)} C^{(i)^T} C^{(i)}$$

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SOLVER

- Optimality of player i

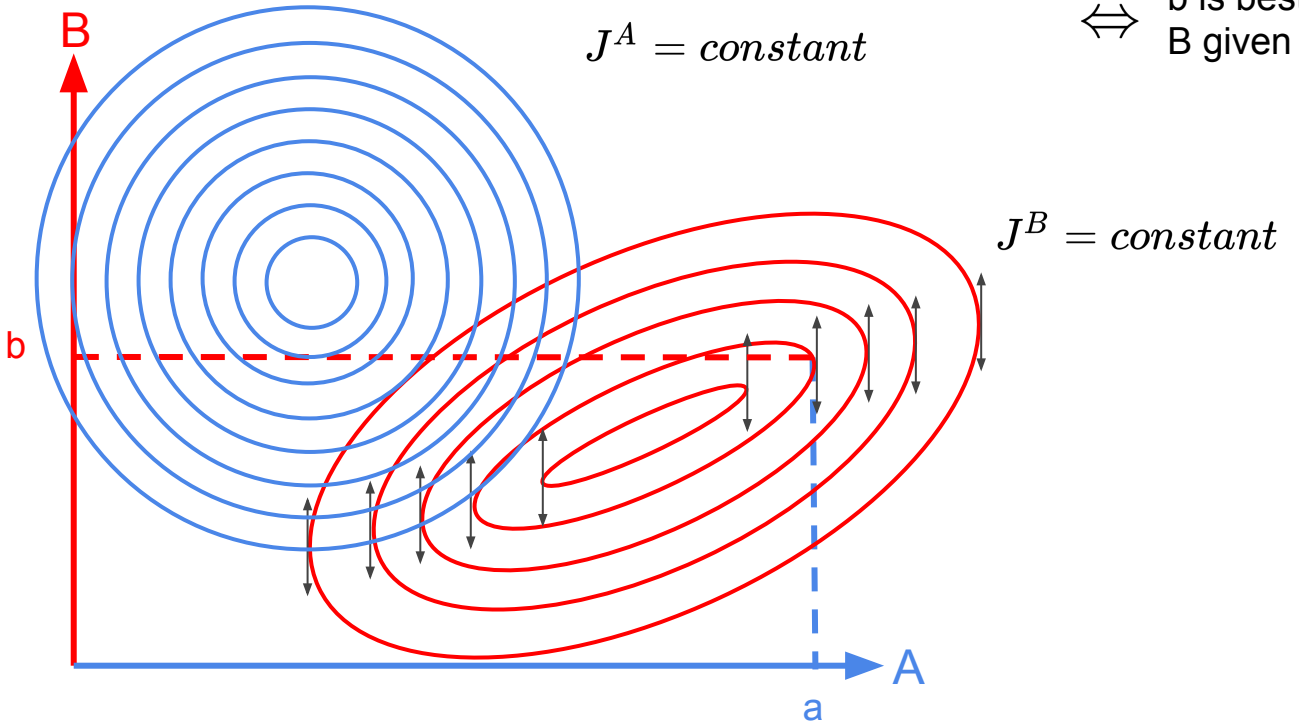
$$\nabla_{P^{(i)}} L^{(i)}(P^{(1)}, P^{(2)}) = 0$$

One constraint for each player.

$$\nabla_{P^{(i)}} L^{(i)} = G^{(i)}$$

DYNAMIC GAMES

Generalized Nash Equilibrium example



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SOLVER

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- Root-finding problem

$$min_{P^{(1)},P^{(2)}} \quad 0$$ No objective

$$s.t. \quad G^{(1)}(P^{(1)}, P^{(2)}) \leq 0$$ Optimality of 1

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- Newton's method

$P = (P^{(1)}, P^{(2)})$ Concatenation

$G = (G^{(1)}, G^{(2)})$ Concatenation

$H = \nabla_P G$ 2nd order derivative

$\delta P = -H^{-1} G$ Search direction

SOLVER

- Trajectory update
  - Line-search
    Find the best step length α in the search direction

$$P_{k+1} = P_k + \alpha\, \delta P_k$$

- Lagrange multiplier update

$$\lambda_{k+1} = \lambda_k + \rho_k C_k$$

- Penalty update

$$\rho_{k+1} = \gamma \rho_k$$

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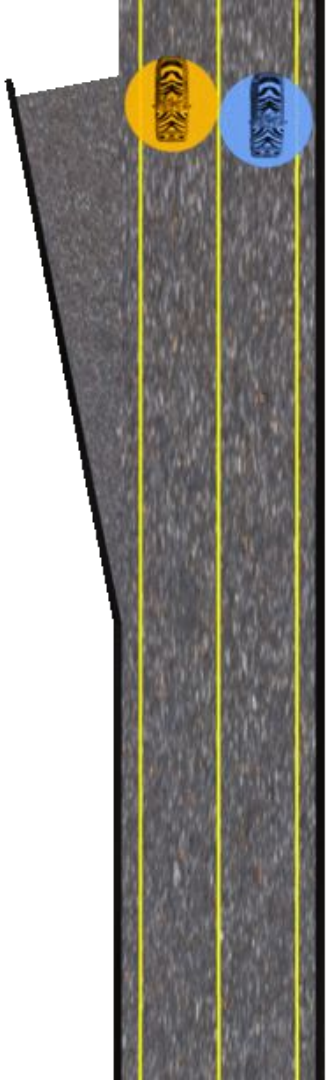
RESULTS

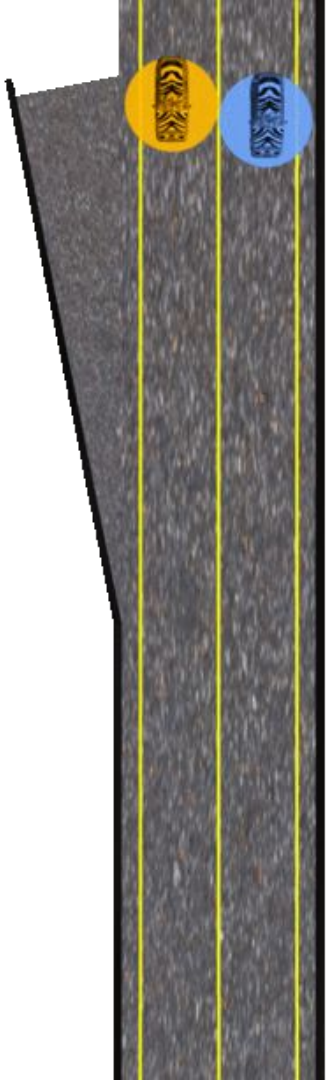
Application to autonomous driving

- 3 types of scenarios:
  - Lane change
  - Overtaking
  - Ramp merging



LANE CHANGE

Successful collision-free lane change



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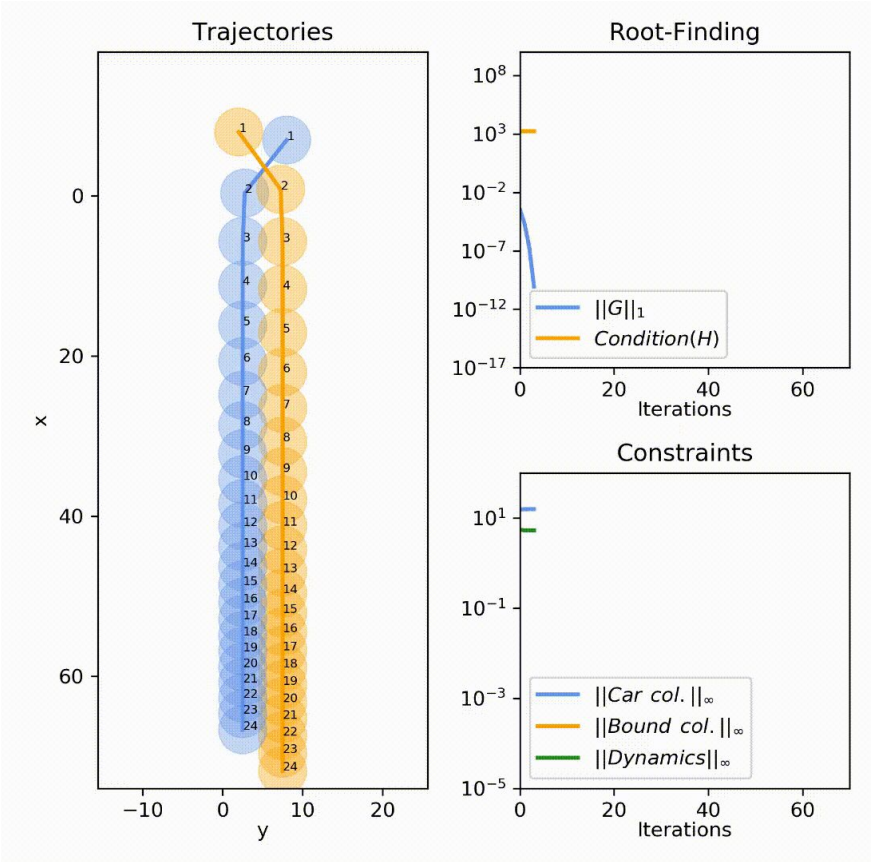
LANE CHANGE

Evolution of the trajectories during the solve.

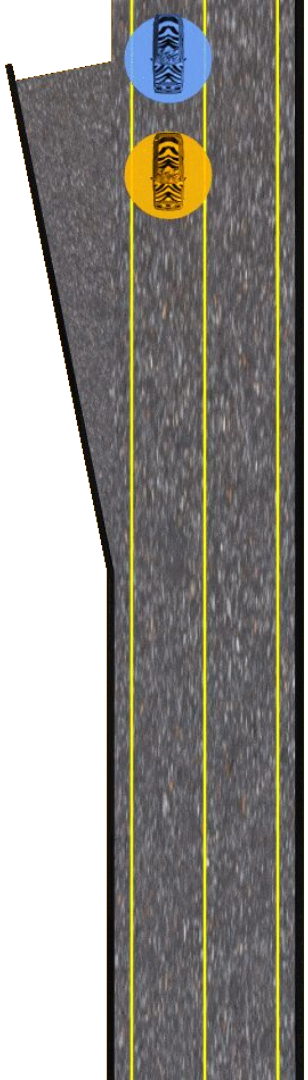
- 2 actors



OVERTAKING

Collision-free trajectories.

- No collision between vehicles
- No collision with black road boundaries



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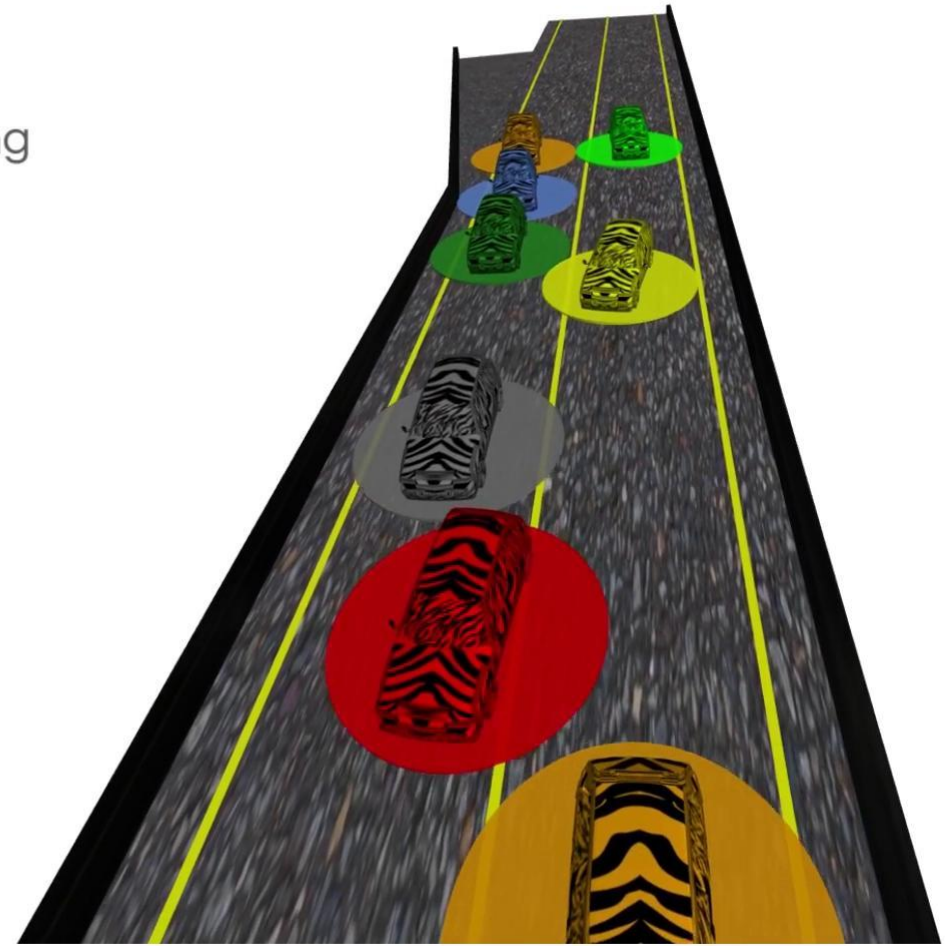

FUTURE WORK

- Parallelization for real-time performance
- Hardware implementation
- Dealing with uncertainty
- Enforcing safety / robustness criteria

ALGAMES

A Fast Solver for Constrained Dynamic Games

GitHub

https://github.com/RoboticExplorationLab/ALGAMES.jl.git

PDF

https://rexlab.stanford.edu/papers/ALGAMES.pdf